TASK-3 (Data Analysis using Pandas)

To demonstrate data analysis using Pandas including data loading, cleaning, filtering, grouping, and aggregation.

Python Code:-

```
import pandas as pd


try:
    # -------- LOAD DATASET --------
    print("Loading dataset...")
    df = pd.read_csv("data.csv")
    print("Dataset loaded successfully.\n")

    # -------- DISPLAY FIRST ROWS --------
    print("Displaying first 5 rows:\n")
    print(df.head())

    # -------- DATASET INFO --------
    print("\nDataset information:\n")
    print(df.info())

    # -------- DATA CLEANING --------
    print("\nCleaning data...\n")

    # Fill missing values in Age column
    df["Age"] = df["Age"].fillna(20)
    print("Missing values in 'Age' column handled.")

    # Remove duplicate rows
    df = df.drop_duplicates()
    print("Duplicate records removed.")

    # Convert Age column to integer
    df["Age"] = df["Age"].astype(int)
```

```
    print("Data type of 'Age' converted to integer.")

    # Select only required columns
    df_simple = df[["Name", "Age"]]
    print("Selected required columns (Name, Age).")

    print("\nData cleaned successfully.\n")

    # -------- FILTERING --------
    print("Filtering data where Age > 20...\n")
    filtered = df_simple[df_simple["Age"] > 20]
    print(filtered)

    # -------- GROUPING --------
    print("\nGrouping data by Name and calculating average Age...\n")
    grouped = df_simple.groupby("Name")["Age"].mean()
    print(grouped)

    # -------- AGGREGATION --------
    print("\nCalculating statistics...\n")
    print("Average Age:", df_simple["Age"].mean())
    print("Maximum Age:", df_simple["Age"].max())
    print("Minimum Age:", df_simple["Age"].min())

    print("\nTask completed successfully.")

except FileNotFoundError:
    print("Error: Dataset file not found. Please check file location.")
except Exception as e:
    print("Unexpected error:", e)
```

Output:-

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Loading dataset...
Dataset loaded successfully.

Displaying first 5 rows:

     Name   Age Department  Salary
0  Pratik  20.0         IT   30000
1   Rahul  22.0         HR   35000
2   Sneha  21.0         IT   32000
3    Amit  23.0    Finance   40000
4    Neha   NaN         IT   31000

Dataset information:

<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Name        8 non-null      str
 1   Age         7 non-null      float64
 2   Department  8 non-null      str
 3   Salary      8 non-null      int64
dtypes: float64(1), int64(1), str(2)
memory usage: 388.0 bytes
None

Cleaning data...

Missing values in 'Age' column handled.
Duplicate records removed.
Data type of 'Age' converted to integer.
Selected required columns (Name, Age).

Data cleaned successfully.

Filtering data where Age > 20...

    Name  Age
1  Rahul   22
2  Sneha   21
3   Amit   23
6  Karan   24
7  Pooja   21

Grouping data by Name and calculating average Age...

Name
Amit      23.0
Karan     24.0
Neha      20.0
Pooja     21.0
Pratik    20.0
Rahul     22.0
Sneha     21.0
Name: Age, dtype: float64

Calculating statistics...

Average Age: 21.571428571428573
Maximum Age: 24
Minimum Age: 20

Task completed successfully.
```

Short Insight Summary:-

The dataset was successfully loaded using Pandas and examined to understand its structure, data types, and missing values. It was observed that the dataset contained a missing value in the Age column and a duplicate record.

Data cleaning was performed by filling missing values and removing duplicate entries to ensure accuracy and consistency. After cleaning, the dataset became more reliable for analysis.

Filtering was applied to extract records where age is greater than 20, which helped in focusing on a specific group of data. Grouping the data by name provided insights into the average age for each individual. Aggregation functions such as mean, maximum, and minimum were used to summarize the data.

Overall, the analysis helped in identifying patterns, improving data quality, and extracting meaningful information from the dataset in a simple and effective way.